

Kısıtlomalar

Okuyucu ve yazıcı
Sığa / sola hareket

turing Makinası

Discrete Dille ve Soyut Makineler

Regüler ifadeler

Grammerler

$\alpha \Rightarrow \beta$

Turing Makinası

Deterministic Finite Automata
(Özdevinir)

Pushdown Automate
(PDA)

$\lfloor z_0 \rfloor$ Yığıt

Turing Makinası

$NP \stackrel{?}{=} P$

a ile b'leri çift uzunluklu katarlar

$$(a+b) ((a+b)(a+b))^* a$$

Pom olarak 2 a içeren dil

$$b^* a b^* a b^*$$

Pom olarak öncelik parenthesisdedir. $\rightarrow$ Öncelik sıratması

Örnek: $L = \{ w \in (a+b)^* \mid n_b(w) = 3k+1 \ k \geq 0 \}$

$$R(L) = ?$$

$k=0$ ise	1 "b" var
$k=1$ ise	4 "b" var
$k=2$ ise	7 "b" var
$k=3$ ise	10 "b" var

1 tane b mutlaka olmalı,

$$a^* b a^* ((b a^*)^3)^*$$

$$\delta^*(q_0, 0) = \bigcap_{s \in \delta^*(q_0, 1)} \delta(q, 0) = \bigcap (\delta(q, 0) \cup \delta(p, 0) \cup \delta(t, 0)) = \{p, u\}$$

$$\delta^*(q_0, 1) = \bigcap (q) = \{q_0, p, t\}$$

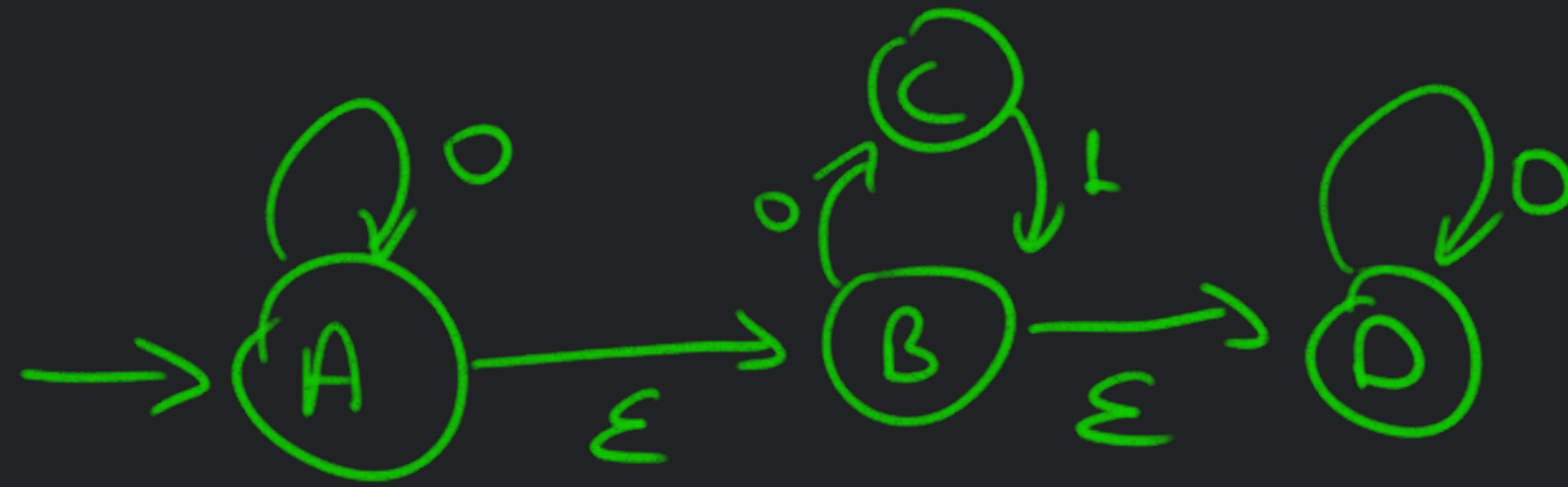
Soru Şöyle ki 5 puan

$$L: \underline{0^*}(\underline{01})^*\underline{0^*} \quad \text{Aynı aynı tanımlar}$$

a: Bunu tanıyan bir NFA-ε

b: NFA

c: DFA



Q \ Σ	δ(q, 1)	δ(q, 0)	δ(q, 1)	δ*(q, 0)	δ*(q, 1)
A	B	A	∅	{A, B, C, D}	∅
B	D	C	∅	{C, D}	∅
C	∅	∅	B	∅	{B, D}
D	∅	D	∅	{D}	∅

$$\{B, D\} \xrightarrow{1} \{\emptyset\}$$

$$\{B, D\} \xrightarrow{0} \{C\}$$

$$\delta^*(A, 0) = \bigcap_{r \in \delta^*(A, \epsilon)} \delta(r, 0) = \bigcap (A, C, D)$$

$\{A, B, D\}$ $\{A, C, D, B\}$

$$\delta^*(A, 1) = \bigcap_{r \in \delta^*(A, \epsilon)} \delta(r, 1) = \bigcap (\emptyset, \emptyset, \emptyset, \emptyset)$$

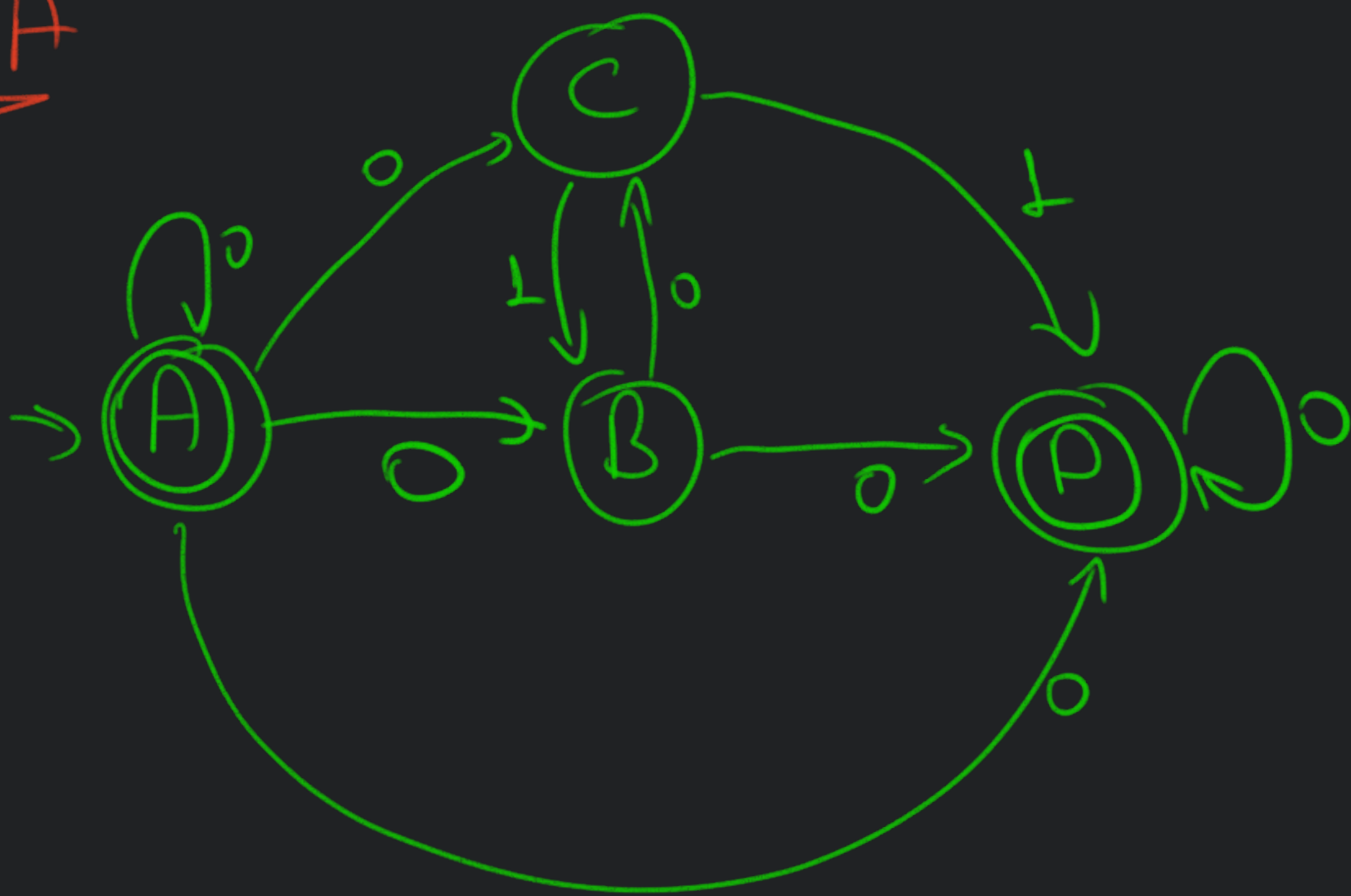
$\{A, B, D\}$ $\{\emptyset\}$

$$\delta^*(B, 0) = \bigcap_{r \in \delta^*(B, \epsilon)} \delta(r, 0) = \bigcap (C, D)$$

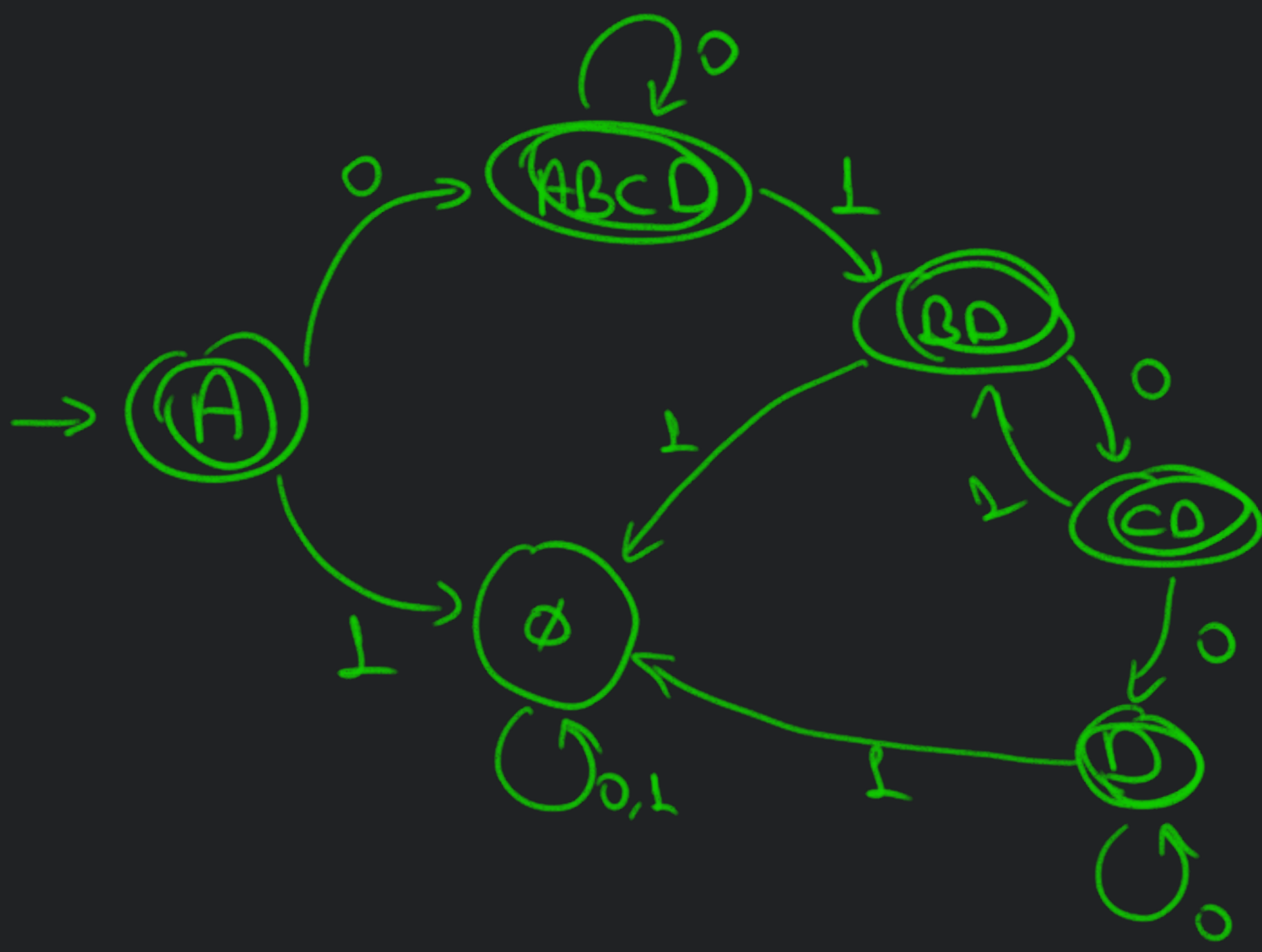
$\{B, D\}$ $\{C, D\}$

NFA

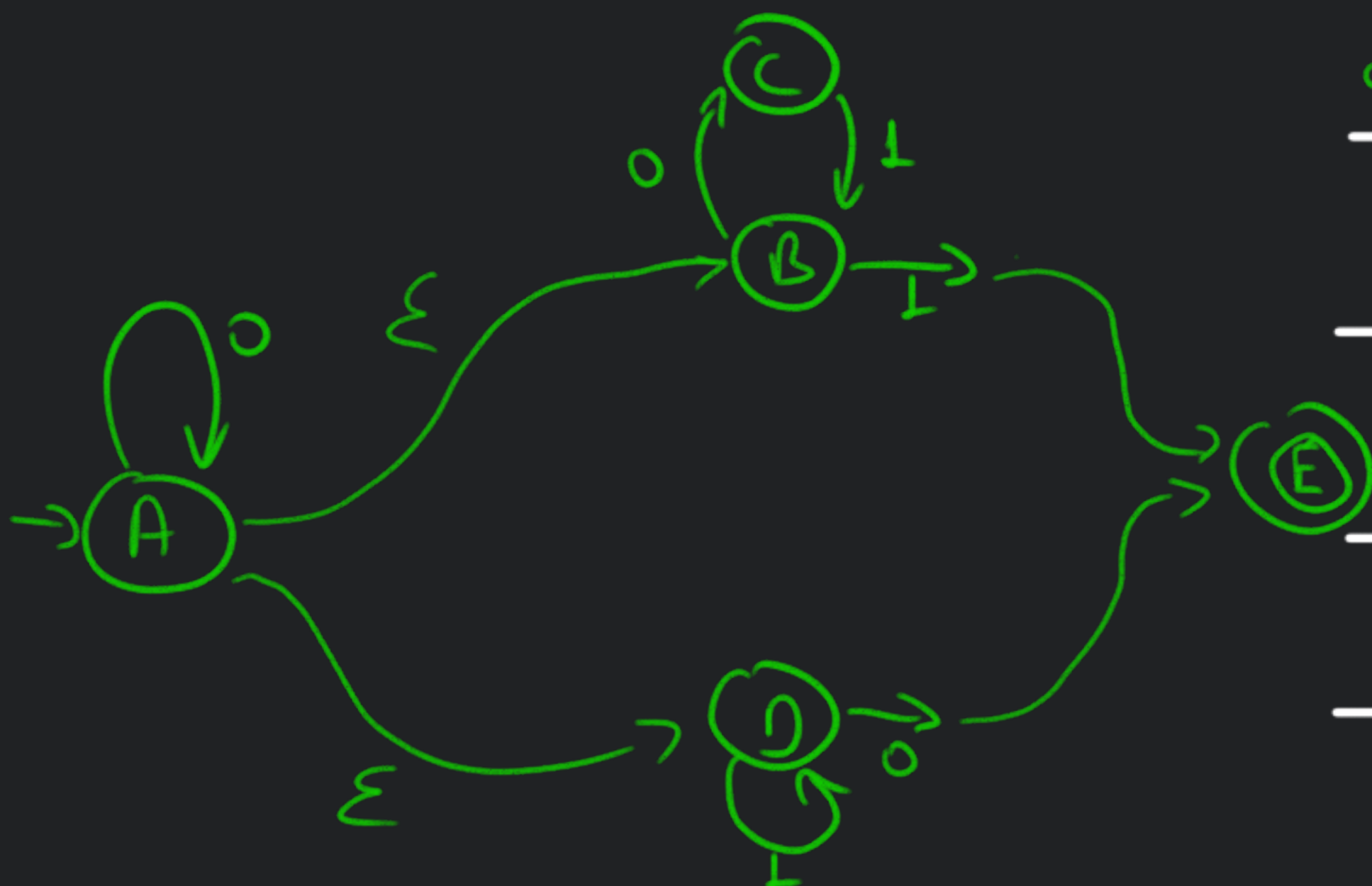
B:



C:

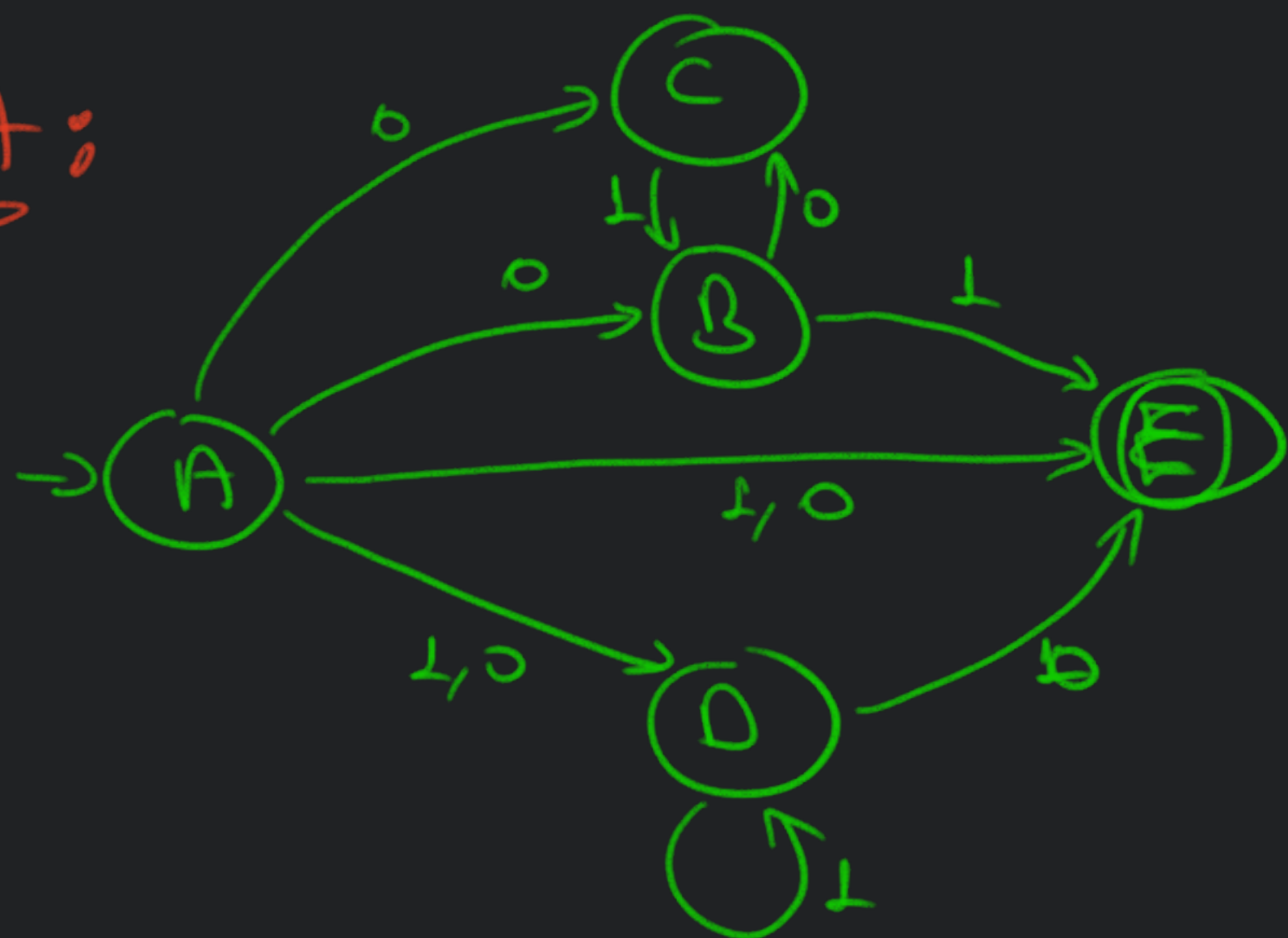


$$L = 0^* ((01)^* 1 + 1^2 0)$$



$q \backslash \Sigma$	$S(q1)$	$S(q0)$	$S(q1)$	$S^*(q0)$	$S^*(q1)$
A	B,D	A	$\emptyset$	A,B,C,D,E	D,E
B	$\emptyset$	C	E	C	E
C	$\emptyset$	$\emptyset$	B	$\emptyset$	B
D	$\emptyset$	E	D	E	D
E	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

NFA:



DFA

